

Student Name:

Daytime: ☐Evening: ☐

Student No :

(20)1-Let $x=2^3+2^{-19}+2^{-22}$ Find the machine numbers on the Marc-32 that are just to the right and just to the left of x .
 (Marc-32 number system: $x=(1.a_1a_2a_3 \dots a_{22}a_{23})_2 * 2^m$ where $-126 \leq m \leq 127$)

(20)2- Find ways to compute the following function without series loss of significance.

$\log(x) - \log(y)$ Assume x and y are positive numbers and x is close to y .

(20)3- Let $x=0.643676895$, and $y=0.643661234$.
 Consider computing $x-y$ using a decimal computer having six-digit mantissa. What will be the relative error in percent?

(20)4-Let $f(x)=x^3-5x^2-x+5$. Perform one iteration of Bisection method (interval halving) on $f(x)$ with starting points $a_0=2$ and $b_0=7$ (Compute $c_0, f(c_0), a_1, b_1, c_1, f(c_1)$,

(20)5-Let $f(x)=x^3-5x^2-x+5$. Perform two iterations of Newton's method on $f(x)$ with starting point $x_0=7$. (compute x_1 and x_2)